$$B_z = \frac{\mu_0}{4\pi}\frac{IR\cdot 2\pi R}{\left(R^2+z^2\right)^{\frac{3}{2}}} = \frac{\mu_0}{2\pi}\frac{I\cdot \pi R^2}{\left(R^2+z^2\right)^{\frac{3}{2}}} = \frac{\mu_0}{2\pi}\frac{p_m}{\left(R^2+z^2\right)^{\frac{3}{2}}}. \quad (3)$$

$z >> R$

$$dB_z = \frac{\mu_0}{2\pi}\frac{p_m}{\left(R^2+z^2\right)^{\frac{3}{2}}} \approx \frac{\mu_0}{2\pi}\frac{p_m}{z^3}, \quad (4)$$

$$b = \frac{\mu_0}{2\pi} \quad (5)$$

$$\beta = 3$$

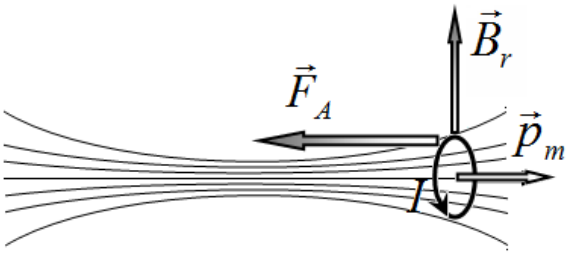
1.1.2

F_A $\vec{B}_r$

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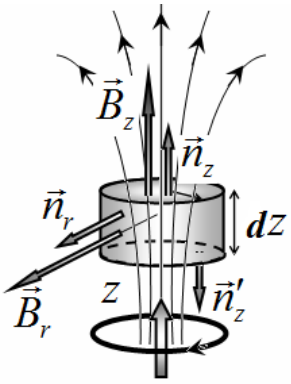
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$$B_z(z+dz)\cdot\pi r^2 - B_z(z)\cdot\pi r^2 + B_r\cdot 2\pi r dz = 0 \quad (6)$$

$$B_r = -\frac{r}{2}\frac{dB_z}{dz} \quad (7)$$

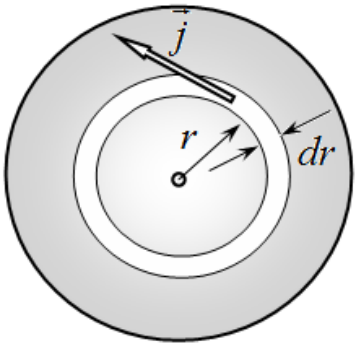


$$F = IB_r 2\pi r = -I\frac{r}{2}\frac{dB_z}{dz}2\pi r = -p_m\frac{dB_z}{dz}. \quad (8)$$

1.2.1

$$\omega_0 = \sqrt{\frac{k}{m}}. \quad (9)$$

1.2.2 .



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(4) .

r dr .

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$$2\pi rE = -\pi r^2\frac{dB_z}{dt} \quad \Rightarrow \quad E = -\frac{r}{2}\frac{dB_z}{dt}. \quad (10)$$